

Total flavonoids of *Rhizoma Drynariae* protect hepatocytes against aflatoxin B1-induced oxidative stress and apoptosis in broiler chickens

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ABSTRACT

Aflatoxin B1 (AFB1) is a common mycotoxin in food and in the environment that lead to multi-organ injury in humans and animals. The objective of this study was to evaluate the detoxification properties of dietary total flavonoids of *Rhizoma drynariae* (TFRD), a Chinese herbal, on aflatoxin B1 (AFB1)-induced hepatic oxidative damage and apoptosis of liver of broiler chickens. A total of 160 healthy specific pathogen free (SPF) 21-day-old broilers were randomly allocated to 4 groups, including the CON group (basal diet), TFRD group (basal diet with 125 mg/kg TFRD), AFB1 group (100 µg/kg body weight), and AFB1 (100 µg/kg body weight) + TFRD (basal diet with 125 mg/kg TFRD) group. The exposure of AFB1 continued for seven days. The results showed that TFRD treatment alleviated the abnormal changes of growth performance and liver morphology, reduced serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST) levels. Moreover, TFRD promoted the antioxidant capacity of serum, increased the activities of total superoxide dismutase (T-SOD), oxidized glutathione (GSSG) and glutathione (GSH) ($p < 0.05$), while decreased MDA contents ($p > 0.05$). Meanwhile, supplementation of TFRD significantly increased the expression of antioxidant-related genes (SOD, CAT, GST, and GPX1) in liver ($p < 0.05$). Furthermore, we found that AFB1 was involved in the regulation of PI3K/AKT signaling pathway, leading to hepatocyte apoptosis. At the same time, TFRD treatment inhibited AFB1-induced apoptosis and significantly changed mRNA expression of apoptosis-related genes, including *PI3K*, *AKT*, *Bax*, and *Bcl-2* ($p < 0.05$). The results indicated that TFRD could alleviate AFB1-induced liver injury in broiler chickens.

1. Introduction

Aflatoxins are secondary fungal metabolites with higher stable properties, mainly produced by *Aspergillus species* of fungi, regarded as highly harmful substances on humans and animals (Zhu et al., 2020; Cao et al., 2021; Li et al., 2021a). Aflatoxins commonly contaminate cereal grains as well as feedstuff (Rushing and Selim, 2019), affects the quality and safety of animal products and threatens human health (Zhao et al., 2021). Aflatoxin B1 (AFB1) exhibits the highest toxicity and the most stable properties and is well-known among the different aflatoxin types, which can lead to hepatotoxicity (Ma et al., 2018), carcinogenicity, teratogenicity, and mutagenicity (Li et al., 2019; Rushing and Selim, 2019; Zhu et al., 2021). Many studies have reported that AFB1 can cause significant economic losses to the poultry and livestock industry. The exposure of AFB1 not only induces liver injury in chickens, ducklings,

sheep, etc. (Liao et al., 2014; Li et al., 2019; Cao et al., 2021) but it's also suppresses the immune system of animals (Zhao et al., 2019), leading to the increased susceptibility of animal infectious diseases. Chickens are particularly sensitive to AFB1 and toxicity occurs mostly due to long-term intake of AFB1-contaminated feed (S. Li et al., 2021; S. Li et al., 2021; Z. Li et al., 2021; X. Li et al., 2021b). AFB1 and its metabolites can accumulate in the broiler liver and cause acute or chronic liver injury or even hepatocellular carcinoma in humans through consumption of contaminated food (Fan et al., 2021; Zhao et al., 2021; Zhu et al., 2021). In poultry, AFB1 exposure can induce oxidative stress, apoptosis, and inflammation resulting in liver injury (Liao et al., 2014; Li et al., 2019; Zhao et al., 2019). Therefore, a safe and effective measure is needed to protect against AFB1-induced liver toxicity.

The previous studies have found that phytochemicals can induce the detoxification of AFB1 through its biotransformation, thereby reducing

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the toxicity of AFB1 (Umayya et al., 2021). Total flavonoids of *Rhizoma drynariae* (TFRD) is a traditional Chinese medicine extracted from the dried root of *Rhizoma Drynariae* herb (Sun et al., 2021). TFRD is widely used bone treatment to treat bone-related diseases due to its effect on strengthening bones and improving bone health (Huang et al., 2020). In recent years, TFRD has attracted much attention due to its pharmacological effects on oxidative stress, inflammation, atherosclerosis, and hyperlipidemia (Kang et al., 2014; Li et al., 2021). Several studies indicated that TFRD significantly elevated the antioxidant capacity in rats with osteoporosis and serum glutathione peroxidase (GPX) activity in laying hens, improving bone health (Huang et al., 2020; Mu et al., 2021). Furthermore, TFRD can reduce ROS generation and inhibit osteocyte apoptosis by activating PI3K/AKT pathway (Lv et al., 2021). In addition, PI3K/AKT signaling pathway also plays an important role in inhibiting the hepatocyte apoptosis (Nie et al., 2021; Wang et al., 2021). However, it is unclear whether TFRD has beneficial effects on AFB1-induced liver injury by regulating oxidative stress and apoptosis.

Therefore, this study was conducted to determine the detoxification effects of dietary TFRD supplementation on AFB1-induced liver injury by detecting liver function indexes, morphological characteristics, oxidation index and apoptosis-related gene expression levels in broiler chickens.

2. Materials and methods

2.1. Chemicals and animals

AFB1 (purity $\geq 99.8\%$, #2J0G25) was obtained from Pribolab Biological Engineering Co., Ltd (Qingdao, China). AFB1 was dissolved in 4% ethanol solution. TFRD (total flavonoids $\geq 95\%$, #K20798) was purchased from Xi'an Kailai Biological Engineering Co., Ltd (Xi'an, China). Assay kits for total superoxide dismutase (T-SOD, #A001-1-2), malondialdehyde (MDA, #A003-1-2), oxidized glutathione (GSSG, #A061-1-2), glutathione (GSH, #A006-2-1), alanine transaminase (ALT, #C009-2-1) and aspartate aminotransferase (AST, #C010-2-1) were

purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, China). Trizol reagent was purchased Takara Biomedical Technology (Beijing) Co., Ltd. (#AKF0728A).

2.2. Experimental design and treatment

A total of 160 three-week-old Arbor Acres (AA) broiler chickens with average initial body weight similarly (838.14 ± 5.64 g) were obtained from a local hatchery (Kaifeng Xingda Poultry Co., Ltd, Kaifeng, China). All chickens consumed basal diets that were formulated according to the Chinese Chicken Feeding Standard (MAPRC, 2004. NY/T33-2004). The broilers were randomly divided into 4 groups with 4 replicates of 10 broilers as per replicate: CON group received basal diet, TFRD group (basal diet with 125 mg/kg TFRD) (Yao et al., 2020), AFB1 group (100 μ g/kg body weight), and AFB1 (100 μ g/kg body weight) + TFRD (basal diet with 125 mg/kg TFRD) group. TFRD was mixed with feed and AFB1 was dissolved in ethanol solution and injected into the ingluviens once a day for seven consecutive days (Fig. 1). The dosage and method of AFB1 and TFRD were based on previous reports (Śliżewska et al., 2019; Yao et al., 2020; Lv et al., 2021). Broilers were offered fresh water, free access to diet, and 12:12 h dark or light cycle. All experiments were approved by the Animal Welfare and Research Ethics Committee of the College of Animal Medicine, Henan Agricultural University (Permit No: 17-0126).

2.3. Sample collection

The daily weight and feed intake of broilers were recorded, to calculate average daily gain (ADG) and average daily feed intake (ADFI). After injection for 7 days, 8 broiler chickens were randomly selected from each group. Broilers were individually weighed, euthanized, and collected for the liver. Blood samples were collected from the jugular vein and centrifuged at 3000g for 15 min at 4 °C to separate serum, was stored at -20 °C for further analysis. The liver samples were weighed, put in liquid nitrogen, and stored at -80 °C for further experiments.

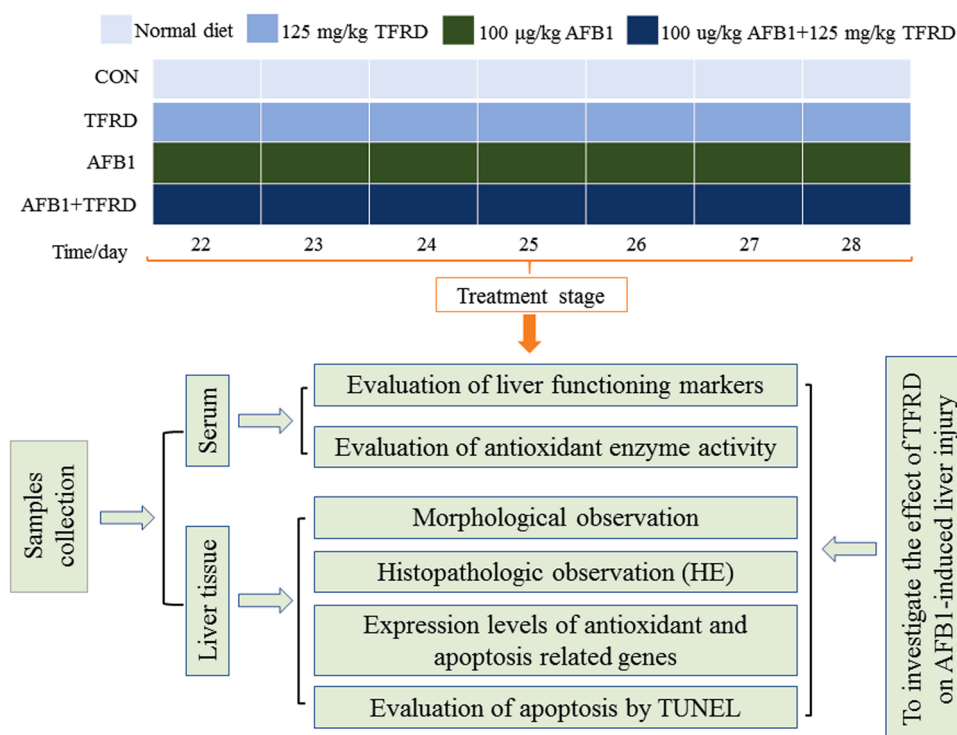


Fig. 1. Schematic diagram of experimental design to study the effect of TFRD on AFB1-induced liver injury in broilers. CON, control group; TFRD, 125 mg/kg TFRD supplement group; AFB1, 100 μ g/kg AFB1 exposed group; AFB1 + TFRD, 100 μ g/kg AFB1 and 125 mg/kg TFRD group.

The relative weight of each liver was calculated by using the formula: liver index (%) = liver weight (g)/body weight (g) × 100.

2.4. Histological examination

The liver tissues were fixed in 4% paraformaldehyde, embedded in paraffin then dewaxed in xylene, dehydrated in graded ethanol solutions and embedded in paraffin. Liver Ssections (4 µm) were stained with hematoxylin and eosin (H&E) (Servicebio Biotechnology Co., Ltd, Wuhan, China), and examined by optical microscopy using a Motic BA600-4 microscope (Motic®, Xiamen, China) to assess liver histopathology.

2.5. Determination of serum biochemical indicators

Serum enzyme activity of liver function-related markers ALT and AST were detected by assay kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China).

2.6. Determination of serum oxidation indices

The serum activities of T-SOD, MDA, GSH and GSSG were detected by using assay kits according to the manufacturer's instructions (Nanjing Jiancheng Bioengineering Institute, Nanjing, China).

2.7. Apoptosis assay by TUNEL

Apoptosis in liver tissues was detected by the TUNEL method (Terminal-Deoxynucleotidyl Transferase Mediated Nick End Labeling). The liver sections (4 µm) were dewaxed in xylene, washed with anhydrous ethanol, repaired with protease K, and washed with balanced buffer PBS for decolorization. The appropriate amount of TDT enzyme and dUTP were incubated at 37 °C for 2 h (TUNEL kit, # G1501, Servicebio Technology Co., Ltd. Wuhan, China). After the nucleus was re-stained with DAPI, the sections were observed under a fluorescence microscope and the images were taken. The positive apoptotic nucleus was green, and the number of apoptotic cells was counted 400 × magnification under an optical microscope. Ortho-Fluorescent microscopy (Nikon Eclipse C1, Japan) and Imaging System (Nikon DS-U3, Japan) were used for apoptotic cells.

2.8. Quantitative real-time PCR (qPCR)

Total mRNA of liver tissues was extracted using the Trizol reagent. The mRNA quality inspection via Thermo Fisher Scientific 5225 Verona Rd (Madison, USA) and then retro-transcription synthesized cDNA. The mRNA sequence of apoptosis-related genes (*PI3K*, *AKT*, *Bax*, and *Bcl-2*), oxidation-related genes (*SOD*, *CAT*, *GST*, and *GPX1*) and *GAPDH* were designed using Primer Premier 5.0 software (Table S1) and synthesized by Tsingke Biotechnology Co., Ltd (Zhengzhou, China). The qRT-PCR was performed according to the SYBR Green I PCR Master Mix (Vazyme Biotech Co., Ltd, China). The PCR amplification thermal cycling consisted of predeformation at 95 °C for 30 s, followed by 40 cycles consisting of denaturation at 95 °C for 10 s, extension at 72 °C for 30 s, and annealing at 60 °C for 30 s. Calculated according to the expression of mRNA using the $2^{-\Delta\Delta CT}$ method. All samples were repeated three times.

2.9. Statistical analysis

All experiments were performed at least three times and representative experiments are demonstrated, unless otherwise stated. The results were performed with the SPSS 26.0 software (SPSS Incorporated, USA) and were analyzed using one-way ANOVA followed by the LSD test as the post-hoc test. Correlation analysis between liver function indexes and oxidation-related parameters or apoptosis-related genes was

performed using Pearson's test, which was performed by the genes cloud tools (a free online platform for data analysis, <https://www.genesccloud.cn>). All experimental data were expressed as mean ± standard error of the mean (SEM), and it was considered that $p < 0.05$ represents statistically significant.

3. Results

3.1. Effect of TFRD on growth performance change in broiler chickens induced by AFB1

As shown in Fig. 2A-C, the effects of TFRD on the growth performance of broilers exposed to AFB1. The ADG of broilers exposed to AFB1 was the lowest as compared with the CON, TFRD and AFB1 +TFRD groups ($p < 0.001$, $p < 0.001$ and $p = 0.003$, respectively). The ADFI of broilers in the TFRD group was the highest ($p < 0.001$), and the ADFI level in the AFB1 group was significantly lower than that in the CON group and AFB1 +TFRD group ($p = 0.018$ and $p = 0.008$, respectively). Consistently, the ratio of feed to weight (ADFI/ADG ratio) in the AFB1 group was significantly higher than that in other groups ($p < 0.001$). These results indicated that TFRD has the potential to alleviate the abnormal growth performance of broilers induced by AFB1.

3.2. Effect of TFRD on AFB1 induced liver injury in broiler chickens

As shown in Fig. 2F-D, the liver weight and liver index of the AFB1 group was significantly higher than that of CON and TFRD groups ($p < 0.001$ and $p < 0.001$, respectively), and after TFRD treatment, which of AFB1 +TFRD group was significantly decreased ($p < 0.001$). In addition, the morphological characteristics of liver tissues were observed and found that the liver tissue was dim and glossless in the AFB1 group (Fig. 2F).

Liver histopathological analysis showed that the liver tissue of the AFB1 group was clearly increased hepatic sinusoid space, poor connectivity between liver cells, karyolysis, cell edema, and blood infiltration (Fig. 2G). TFRD significantly alleviated abnormal liver changes and restored liver injury under the AFB1 exposure. In addition, compared with the CON group, serum ALT and AST activities of broilers exposed to AFB1 were significantly increased ($p < 0.001$ and $p < 0.001$, respectively; Fig. 2H-I). Compared with the AFB1 group, the ALT and AST levels in the AFB1 +TFRD group showed a significant suppressed ($p < 0.001$ and $p < 0.001$, respectively). These results demonstrated that TFRD can reverse the abnormal changes of serum biochemical parameters and protect against liver injury induced by AFB1.

3.3. Effect of TFRD on AFB1-induced oxidative damage in broiler chickens

The contents of serum T-SOD, MDA, GSH and GSSG were detected to evaluate the redox status of broiler. As shown in Fig. 3A-E, compared with the CON group, the serum T-SOD, GSH and GSSG activities of broilers in the AFB1 group were significantly decreased ($p < 0.001$, $p < 0.001$ and $p = 0.01$, respectively), while the GSH/GSSG ratio showed obviously increased ($p = 0.031$) and MDA activity upregulated to 9.55% ($p > 0.05$). Subsequently, we found that TFRD can reverse the abnormal changes in oxidation indexes caused by AFB1. For example, the activities of T-SOD, GSH and GSSG in the AFB1 +TFRD group were significantly elevated ($p < 0.001$, $p < 0.001$ and $p = 0.001$, respectively) compared with the AFB1 group, and returned to a similar level as those in the CON group. While the activity of MDA in the AFB1 +TFRD group was also decreased 18.86% compared with the AFB1 group ($p = 0.037$).

The expression level of oxidative stress-related gene mRNA in liver was consistent with the change in serum detection level (Fig. 3G-J). Compared with the CON group, the expression levels of *SOD*, *CAT*, *GST* and *GPX1* genes in the AFB1 group were significantly downregulated

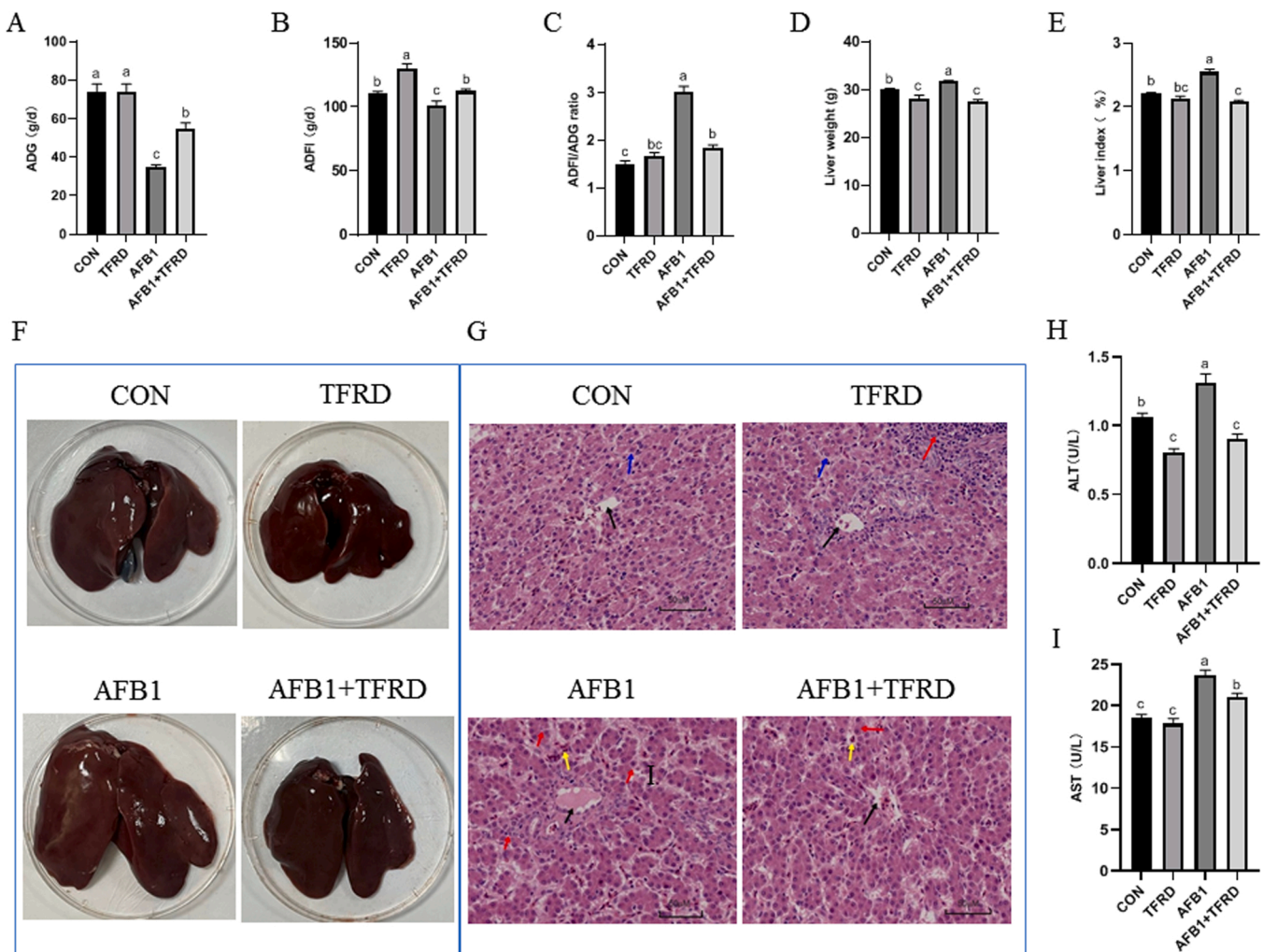


Fig. 2. Effects of TFRD on AFB1-induced growth performance, liver tissue morphology, structure and functional abnormalities in broilers. (A-C) Growth performance related parameters of broilers: average daily gain (ADG); average daily feed intake (ADFI); ratio of feed to weight (ADFI/ADG), respectively. (D-E) liver weight and liver index. (F) Hepatic morphology. (G) Liver tissue HE results (200 ×): CON group and TFRD group, liver cells (blue arrow) are radially arranged around the central vein (black arrow) to form hepatocyte cords, and lymphocytes are abundant in lobules (red arrow). AFB1 group, central vein (black arrow), hepatic cell edema, karyolysis, and partial nuclear fragmentation (red arrow); There is blood infiltration in the sinus space (yellow arrow). AFB1 +TFRD group, closely packed hepatocytes with reduced karyolysis (red arrow); the area of hemo infiltration is reduced (yellow arrow). (H-I) The activities of ALT and AST in serum. ALT, alanine aminotransferase; AST, aspartate aminotransferase. CON, control group; TFRD, 125 mg/kg TFRD supplement group; AFB1, 100 µg/kg AFB1 exposed group; AFB1 +TFRD, 100 µg/kg AFB1 and 125 mg/kg TFRD group. The results are represented as mean ± SEM. a, b and c represent significant differences between groups ($p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

($p = 0.008$, $p < 0.001$, $p = 0.035$ and $p = 0.015$, respectively). The mRNA expressions of *SOD*, *CAT*, *GST* and *GPX1* in the AFB1 +TFRD group were significantly higher than those in the AFB1 group ($p < 0.001$), which were similar to those in the CON group. In addition, oxidative stress-related genes showed higher expression levels in the TFRD group, suggesting that TFRD can significantly improve the antioxidant capacity of the broiler liver.

Pearson correlation analysis was used to explore the correlation between antioxidant indexes and liver function-related parameters (Fig. 3K). The results showed that the liver function-related parameters were negatively correlated with the activities of serum antioxidant enzymes T-SOD, GSSG and GSH, but positively correlated with MDA activity. Liver antioxidant genes *SOD*, *CAT*, *GST* and *GPX1* were negatively correlated with the liver function-related parameters. Among them, liver index was significantly correlated with the activities of T-SOD, GSSG and GSH ($r = -0.728$, -0.761 and -0.744 , $p = 0.007$, 0.004 and 0.005 , respectively). The serum ALT activity was significantly correlated with the serum levels of T-SOD, CAT, and GST ($r = -0.718$, -0.594 and -0.671 , $p = 0.008$, 0.042 and 0.017 , respectively). The serum AST

activity was significantly correlated with serum GSSG, GSH, SOD, CAT and GPX1 levels ($r = -0.637$, -0.762 , -0.618 , -0.732 and -0.614 , $p = 0.026$, 0.004 , 0.032 , 0.007 and 0.033 , respectively). These correlation indexes indicated that liver function is closely related to antioxidant indexes.

3.4. Effect of TFRD on AFB1 induced apoptosis of broiler hepatocytes

As shown in Fig. 4A, TUNEL staining results of broiler liver tissue. Compared with the CON group, the positive cell rate of hepatocytes in the AFB1 group was significantly increased ($p = 0.001$), while there was almost no difference in the positive cell rate of hepatocytes in the TFRD group ($p = 0.446$; Fig. 4B). In addition, compared with AFB1 group, the addition of TFRD to broiler feed exposed to AFB1 can significantly reduce the rate of hepatocyte positive cells ($p = 0.002$). The findings indicated that the addition of TFRD in feed can inhibit AFB1-induced hepatocyte apoptosis.

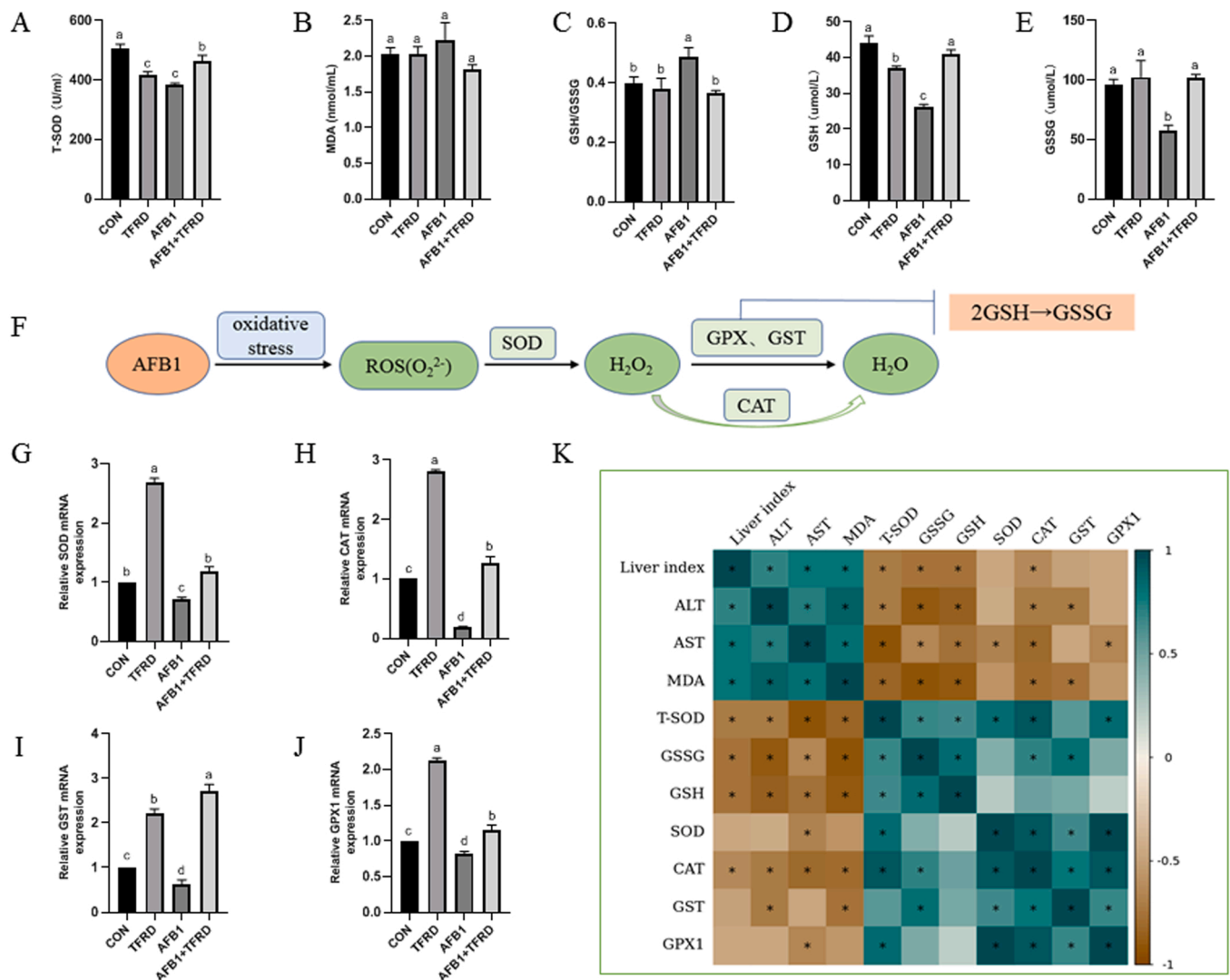


Fig. 3. Effect of TFRD on AFB1-induced liver oxidative damage in broilers. (A-E) Biochemical analysis of serum T-SOD, MDA, GSH and GSSG levels, and GSH/GSSG ratio. (F) ROS production and scavenging. (G-J) The relative mRNA expressions of oxidative stress related genes *SOD*, *CAT*, *GST* and *GPX1* in liver. (K) Pearson correlation analysis between antioxidant indexes and liver function-related parameters. T-SOD, Total superoxide dismutase; MDA, malondialdehyde; GSH, glutathione; GSSG, oxidized glutathione; GST, glutathione-S-transferase; GPX1, glutathione peroxidase 1. CON, control group; TFRD, 125 mg/kg TFRD supplement group; AFB1, 100 μg/kg AFB1 exposed group; AFB1 +TFRD, 100 μg/kg AFB1 and 125 mg/kg TFRD group. The results are represented as mean ± SEM. a, b and c represent significant differences between groups ($p < 0.05$).

3.5. Changes in PI3K/AKT signaling pathway related gene expression

To evaluate the role of PI3K/AKT signaling pathway in TFRD against AFB1-induced hepatocyte apoptosis, we detected the mRNA expression of genes *PI3K*, *AKT*, *Bax* and *Bcl-2* (Fig. 5A-D). The results showed that the mRNA expression of the anti-apoptotic gene *Bcl-2* was significantly down-regulated ($p < 0.001$), while *Bax* mRNA expression was significantly up-regulated ($p < 0.001$) in the AFB1 group compared with the CON group. In addition, the mRNA expressions of *PI3K* and *AKT* in the AFB1 group were decreased 26.28% and 19.23% than those in the CON group respectively ($p > 0.05$ and $p > 0.05$). Compared with the AFB1 group, the mRNA expression levels of *PI3K*, *AKT* and *Bcl-2* genes were significantly up-regulated ($p = 0.003$, $p = 0.001$ and $p < 0.001$, respectively), while *Bax* gene was significantly down-regulated ($p < 0.001$) in the AFB1 +TFRD group.

Pearson correlation analysis was further used to explore the correlation between the gene expression levels of PI3K/AKT signaling pathway and liver function-related parameters (Fig. 5E). Liver function-related parameters with the gene expression level of pro-apoptotic gene

Bax was positively correlated, while negatively correlated with anti-apoptotic gene *Bcl-2*. In addition, the gene expression levels of *PI3K* and *AKT* were negatively correlated with liver function-related parameters. Among them, the *PI3K* gene was significantly negatively correlated with serum AST activity ($r = -0.659$, $p = 0.020$). The *Bcl-2* gene was significantly correlated with liver index, serum AST and ALT levels ($r = -0.707$, -0.772 and -0.821 , $p = 0.010$, 0.003 and 0.001 , respectively). whereas the *Bax* gene was significantly positively correlated with liver index ($r = 0.772$, $p = 0.003$). These data indicated that AFB1-induced hepatocyte apoptosis through PI3K/AKT signaling pathway, which is closely related to liver dysfunction.

4. Discussion

Long-term consumption of AFB1 contaminated feed by livestock and poultry will form a certain accumulation in the body, and then AFB1 toxin residues in livestock and poultry products will cause serious health and economic harm to itself and even human beings (Fan et al., 2013; Saleemi et al., 2020). Controlling feed mildew and alleviating or

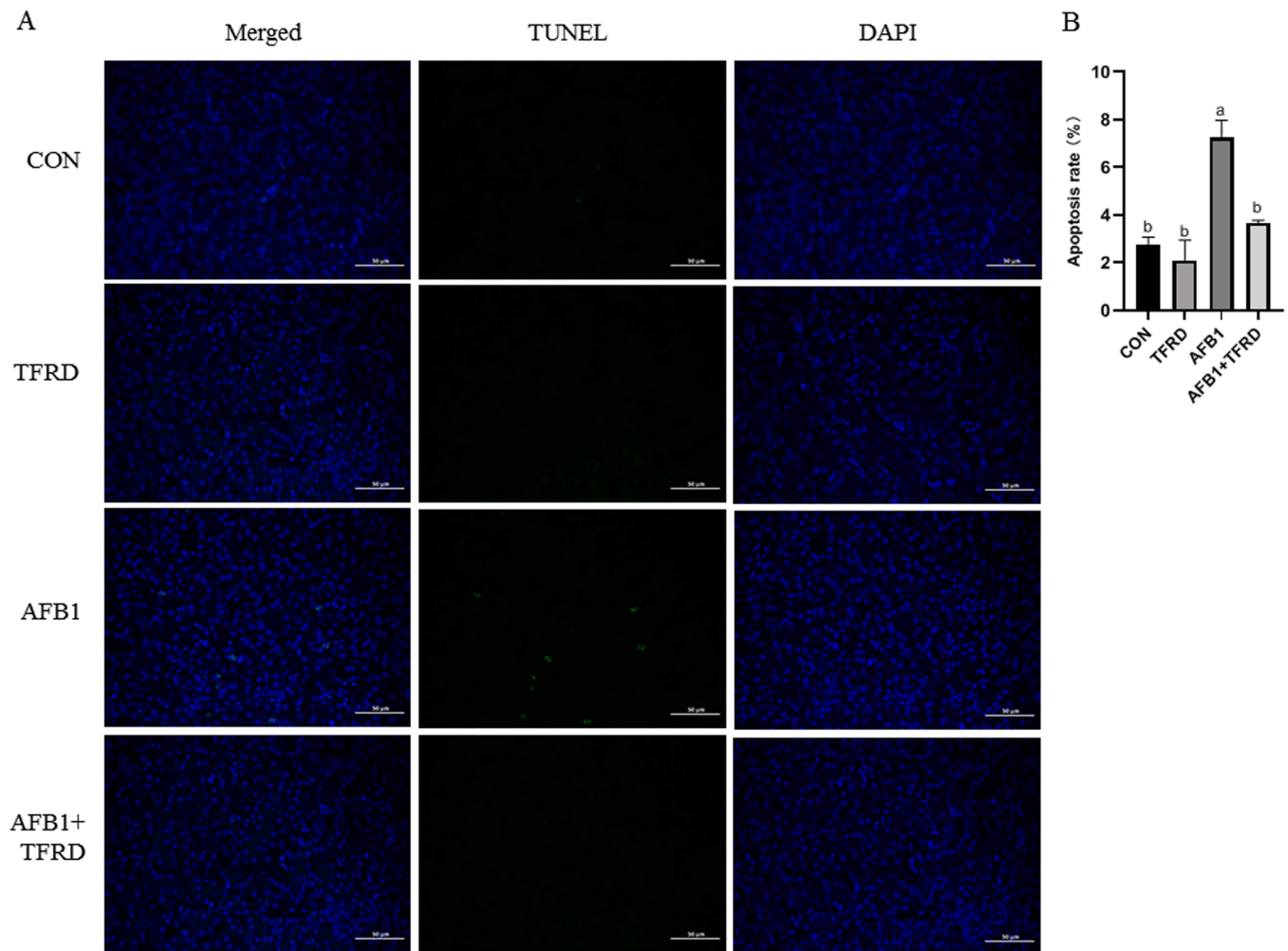


Fig. 4. Effect of TFRD on AFB1-induced hepatocyte apoptosis of broilers. (A) TUNEL staining results of broiler liver tissue, green fluorescence showed TUNEL positive cells, DAPI staining for nuclear (400 ×). (B) Percentage of TUNEL positive cells. CON, control group; TFRD, 125 mg/kg TFRD supplement group; AFB1, 100 ug/kg AFB1 exposed group; AFB1 +TFRD, 100 ug/kg AFB1 and 125 mg/kg TFRD group. The results are represented as mean ± SEM. a, b and c represent significant differences between groups ($p < 0.05$).

inhibiting AFB1 poisoning in livestock and poultry have always been key issues in agricultural development. Inhibition of oxidative stress-induced liver damage by phytochemicals supplementation has become an attractive reasonable strategy for the prevention and treatment of liver-related diseases (A V et al., 2017; Li et al., 2020). Our study confirmed that AFB1 inhibited the growth performance of broiler chickens and had abnormal effects on the liver antioxidant system and hepatocyte apoptosis (Xu et al., 2017; Saleemi et al., 2020), while supplementation of TFRD in feed can reverse the adverse effects in the liver caused by AFB1.

The decrease in feed intake, body weight gain, and feed conversion rate of poultry is one of the characteristics of AFB1 poisoning (Elwan et al., 2021). In the present study, we noticed that AFB1 exposure reduced ADG and ADFI, resulting in growth retardation of broilers, which may be due to anorexia, thirst and mental retardation caused by AFB1. Our findings are consistent with the study by Tsiouris et al. (2021). The liver is considered the main target organ of AFB1 poisoning. We found the exposure of AFB1 causes increase in liver index, pathogenic lesions liver surface, and degeneration of the hepatocytes have been observed indicating AFB1-triggered liver injury. Commonly, the changes of serum ALT and AST activity levels are clinical markers for evaluating liver function (Chen et al., 2021). Relevant studies have confirmed that the ALT and AST activities in serum of AFB1-exposed broilers were increased, which is consistent with the findings of our

previous study (X. Li et al., 2021; Z. Li et al., 2021; S. Li et al., 2021). It is indicated that short-term exposure to high concentration of AFB1 could cause liver damage to broiler chickens. In addition, the improvement effect of TFRD on the growth performance of broilers is also traceable (Lv et al., 2021). Studies on mouse liver also showed that feeding some plant extracts such as lycopene regulated ALT and AST levels through blood circulation, thereby preventing liver function from AFB1 damage (Xu et al., 2017). Our results demonstrated that feeding the dietary antioxidants as TFRD can alleviate AFB1-induced liver injury in livestock and poultry.

Previous studies have confirmed that aflatoxin or other mycotoxins damage the liver is mainly caused by oxidative damage of liver cells, damage the antioxidant system of the liver, resulting in liver dysfunction (Kövesi et al., 2019; Xu et al., 2020; Wan et al., 2021). The liver is the main part of glutathione biosynthesis, and the antioxidant system of the body is jointly regulated by antioxidant enzymes and glutathione redox system (Guengerich et al., 1996; Shendge et al., 2021). In the present study, it was noticed that the liver damage caused by AFB1 inhibits serum antioxidant enzyme activity, meanwhile elevates oxidative stress parameter level and GSH/GSSG ratio, resulting in the disorder of liver redox system. It is worth noting that the GSH level in serum decreased, and the activity level of GSSG also decreased, which is inconsistent with the results of Cheng et al. (2019). Therefore, we speculated that AFB1 exposure in this experiment led to a decrease in serum total glutathione

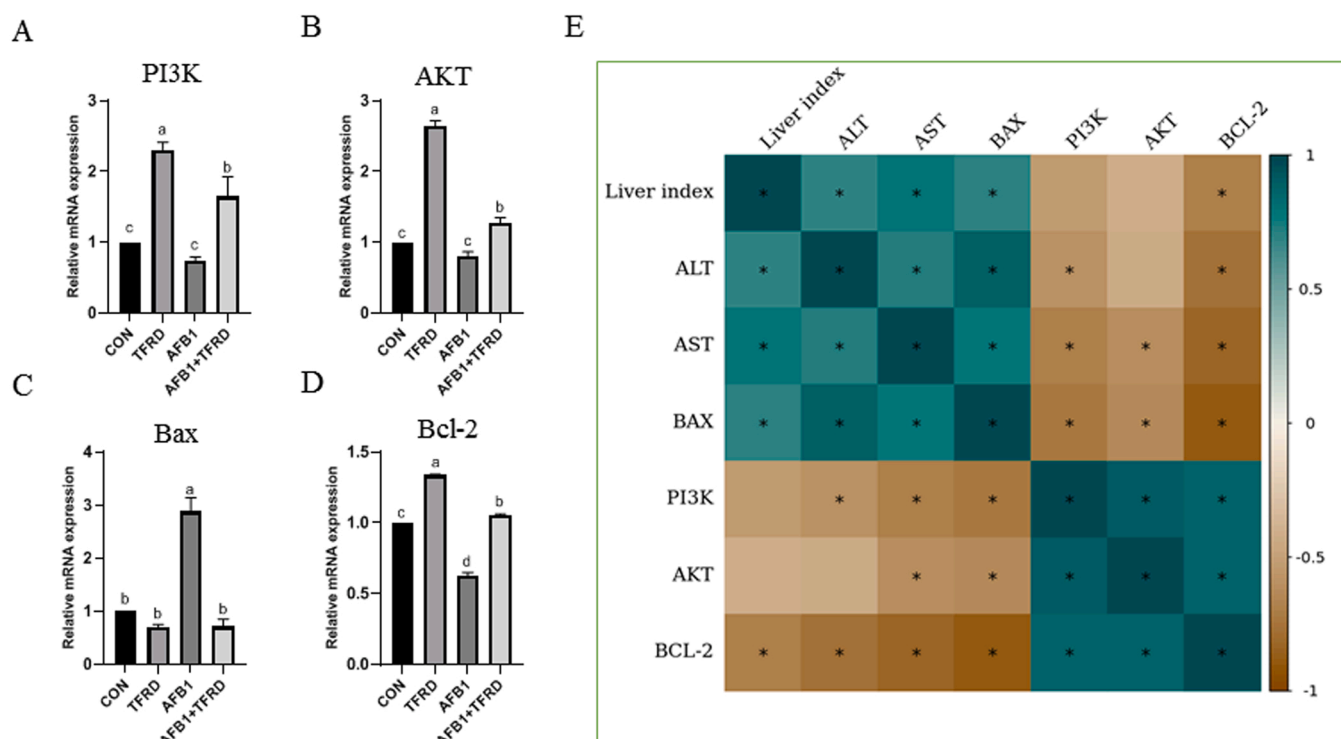


Fig. 5. Gene expression of apoptosis-related pathways. (A-B) The mRNA expression of *PI3K* and *AKT*. (C-D) The mRNA expression of *Bax* and *Bcl-2*. (E) Pearson correlation analysis between apoptosis genes and liver function-related parameters. CON, control group; TFRD, 125 mg/kg TFRD supplement group; AFB1, 100 ug/kg AFB1 exposed group; AFB1 +TFRD, 100 ug/kg AFB1 and 125 mg/kg TFRD group. The results are represented as mean $\pm$ SEM. a, b and c represent significant differences between groups ($p < 0.05$).

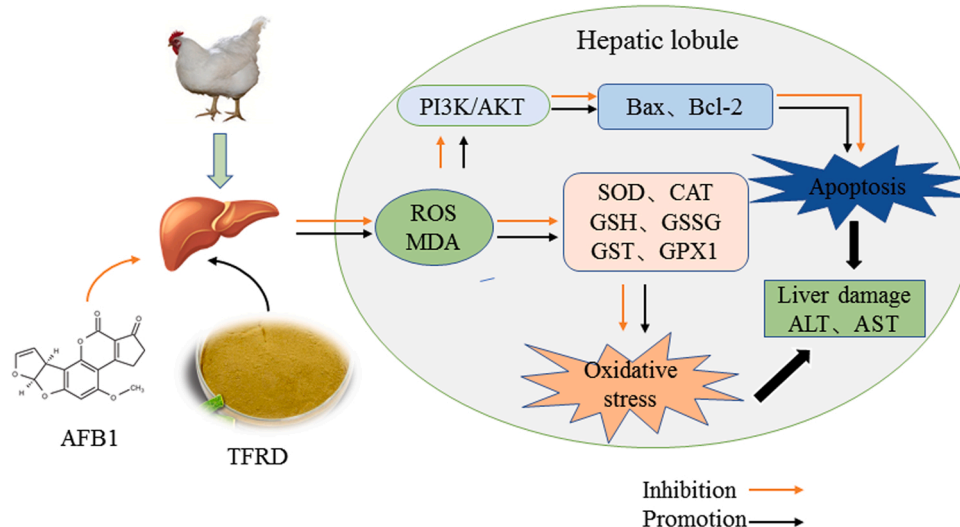


Fig. 6. The mechanism of TFRD on AFB1-induced liver oxidative stress and apoptosis.

synthesis in broilers. In addition, after comparison with previously published data, we confirmed that TFRD, a natural antioxidant and free radical scavenger, could be able to increase the anti-oxidative ability and regulate the glutathione redox system of the broiler liver and to alleviate the AFB1-induced oxidative stress (Xu et al., 2020; Mu et al., 2021; Zhao et al., 2021).

AFB1 can induce oxidative damage in the liver by producing excessive reactive oxygen species and oxygen free radicals, which can result in hepatocyte apoptosis (Fan et al., 2021). In addition, glutathione depletion is sufficient to induce apoptosis of the cell system, especially in hepatocytes (Meki et al., 2004). Our result showed that the apoptotic

rate of hepatocytes by TUNEL method in the AFB1 group was significantly increased compared with the CON group, indicating hepatocyte apoptosis caused by AFB1 exposure. This result matches the previous studies, further confirming that AFB1 can result in hepatocyte oxidative damage and even apoptosis (Liao et al., 2014; X. Li et al., 2021; S. Li et al., 2021; S. Li et al., 2021; Z. Li et al., 2021; Rajput et al., 2021). Traditional Chinese medicine and its extracts have a good effect on improving the AFB1-induced liver apoptosis. Tao et al. (2021) pointed out that *Penthorum chinense Pursh* compound ameliorates AFB1-induced oxidative stress and apoptosis via modulation of mitochondrial pathways in broiler chicken kidneys. Similar findings in our experiment also

show that TFRD have a good effect on improving the AFB1-induced liver apoptosis.

In addition, some studies have confirmed that PI3K/AKT signaling pathway is an important pathway involved in regulating cell growth and metabolism, cell apoptosis and autophagy, which is also a targeted pathway for intervention and treatment (Song et al., 2021; Teng et al., 2021; Wang et al., 2021; Yang et al., 2021). Huang et al. (2019) reported that AFB1 exposure down-regulated PI3K and AKT protein expression. Moreover, a study by Yang et al. (2013) pointed out that activation of PI3K/AKT pathway could neutralize AFB1-induced hepatocyte apoptosis, which is consistent with our results. *Bax* and *Bcl-2* are a pair of antagonistic genes in the *Bcl-2* gene family, and their gene or protein expression can reflect cell apoptosis (Zhang et al., 2019; Li et al., 2021). Moreover, upregulation of phosphorylated AKT suppresses the release of *Bax*, thereby activating the downstream *Bcl-2* to inhibit cell apoptosis (Song et al., 2021). In the present study, our results revealed that exposure of AFB1 had up-regulated the expression of *Bax* and down-regulated the expression of *Bcl-2*, *PI3K* and *AKT*, suggesting the PI3K/AKT signaling pathway involved possibly in AFB1-induced hepatocyte apoptosis in broiler. In addition, the expression levels of *PI3K* and *AKT* genes were up-regulated after TFRD supplementation in feed. It can be seen that TFRD may promote hepatocyte proliferation and inhibit apoptosis by activating PI3K/AKT pathway (Lv et al., 2021).

Flavonoids are found in grains, roots, stems, flowers, fruits, vegetables and tea etc., a kind of natural substances with varied phenolic structures (Panche et al., 2016). Related studies have found that some flavonoids can regulate cellular activity and fight off free radicals that cause oxidative stress and cell apoptosis (G. Yang et al., 2021; C. Yang et al., 2021). The supplementation of luteolin could alleviate AFB1-induced apoptosis and oxidative stress in the liver of mice through activation of the Nrf2 signaling pathway (Rajput et al., 2021). Lv et al. (2021) demonstrated that TFRD can up-regulate the protein expression levels of PI3K and AKT, reverse the changes of *Bax* and *Bcl-2* expression. Our results showed that the supplement of TFRD in feed can regulate the expression of apoptosis-related genes and inhibit the apoptosis of broiler liver cells. Based on the above results, we conclude that TFRD can alleviate AFB1-induced liver damage through inhibiting the oxidative stress and anti-apoptosis signaling pathway in broiler chickens. Fig. 6.

5. Conclusion

In summary, our study confirmed that oxidative stress and apoptosis were involved in AFB1-induced liver injury and liver dysfunction in broiler chickens. Dietary TFRD improved growth performance of broilers, up-regulated T-SOD, GSH and GSSG activities in serum, as well as the expression of antioxidant-related genes, and inhibited the formation of MDA and the expression of pro-apoptotic gene *Bax*, thus protecting broilers from AFB1-induced oxidative stress and apoptosis. This study first illustrated the alleviation effect of TFRD on AFB1-induced oxidative stress and hepatocyte apoptosis in broiler chickens, which provide a valuable reference for further exploring the toxic effects of AFB1 on poultry and the antagonistic mechanism of TFRD on AFB1.

CRediT authorship contribution statement

Luxi Lin: Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. **Pengfei Fu:** Methodology, Data curation, Investigation. **Nuoman Chen:** Methodology, Investigation. **Ningyu Gao:** Investigation. **Qinqin Cao:** Investigation. **Ke Yue:** Methodology. **Tingting Xu:** Methodology. **Chaodong Zhang:** Investigation. **Xilan Jiao:** Supervision. **Cai Zhang:** Supervision. **Fang Liu:** Supervision. **Xuebing Wang:** Supervision. **Shucheng Huang:** Conceptualization, Supervision, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2021.113148.

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